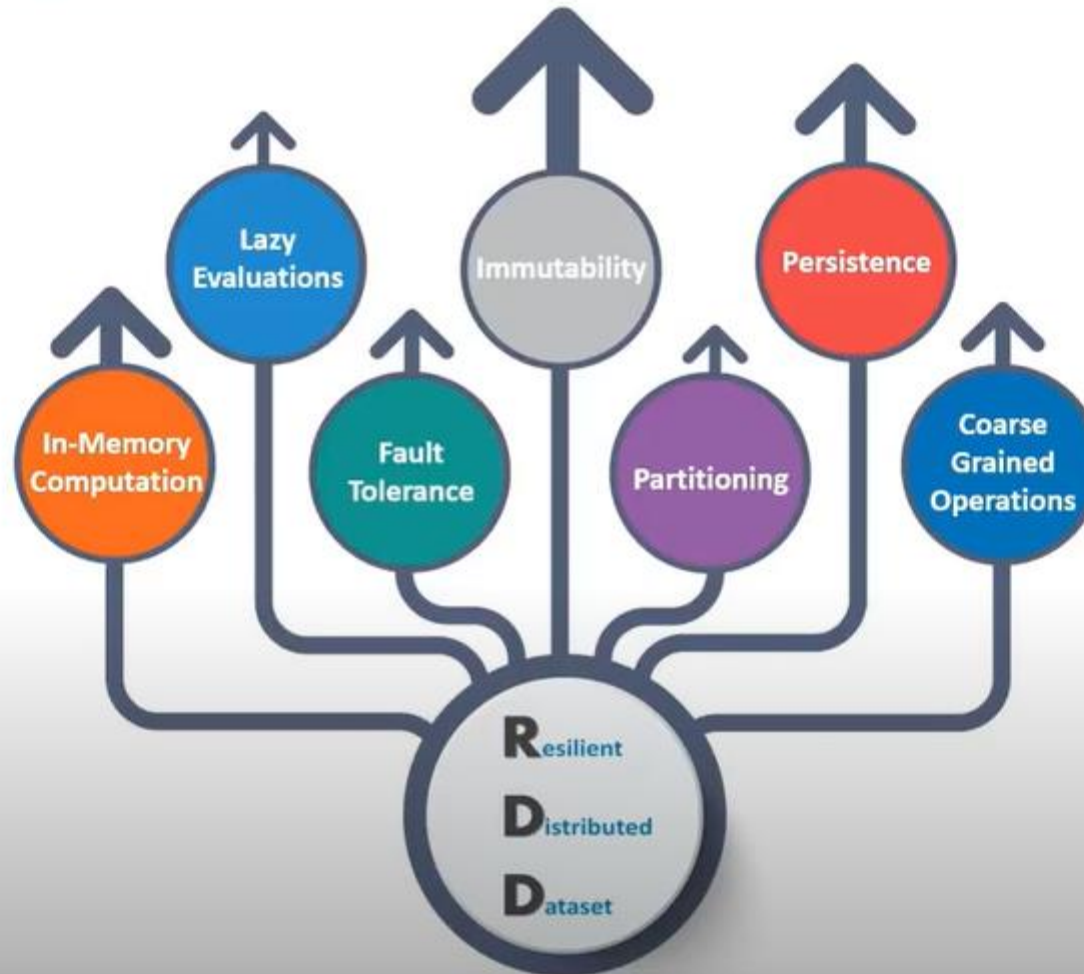
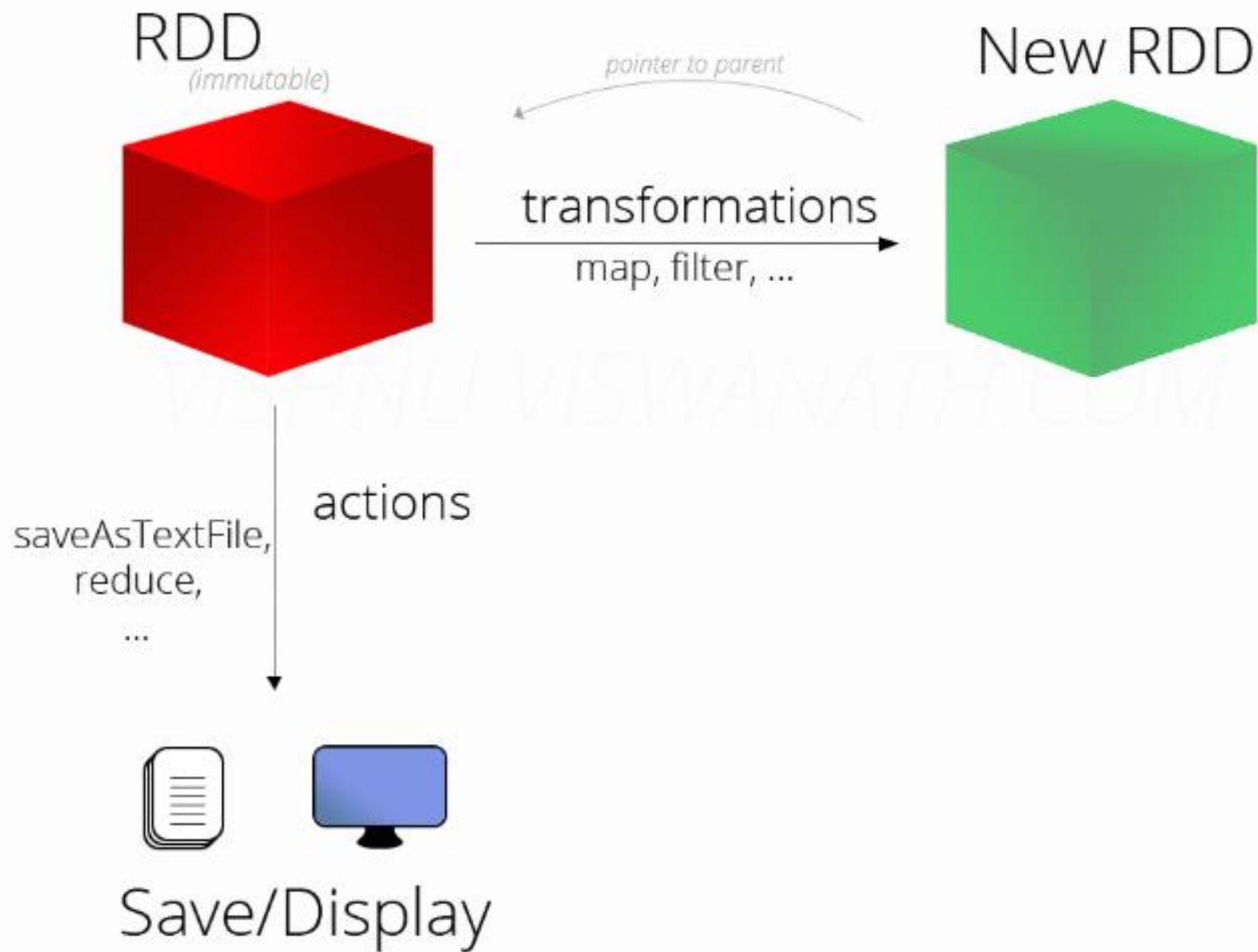


Introduction

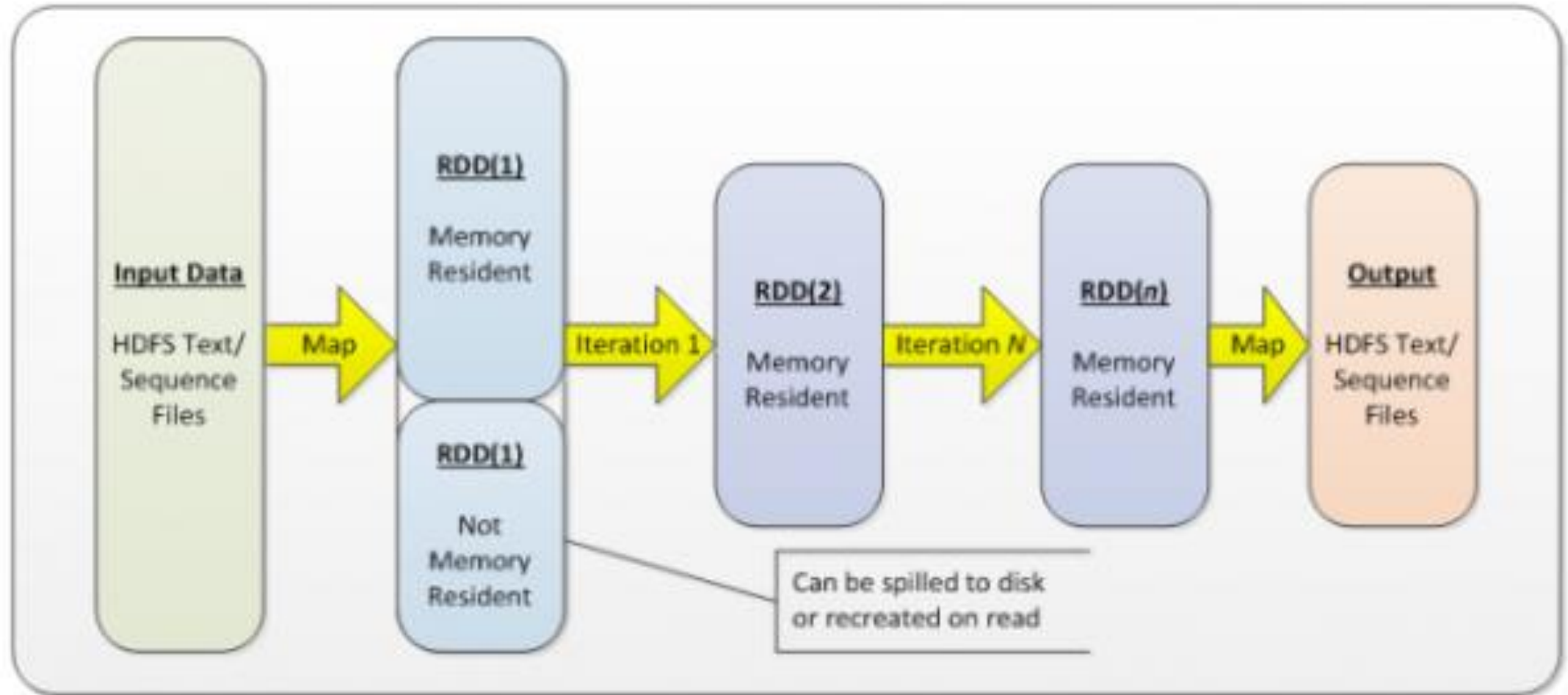
- RDD represents Resilient Distributed Dataset. It is immutable, distributed and contains a collection of object sets
- RDD can be transformed into another RDD
- A string of RDDs established a pipeline that can be executed in a Spark system using “Lazy execution” philosophy.
- RDDs may be computed on different worker nodes in a cluster
- This module contains both RDD transformations and actions

Features of RDDs





RDD transformations pipeline



A Word Count Example using RDD



Spark: Visualizing RDD transformations

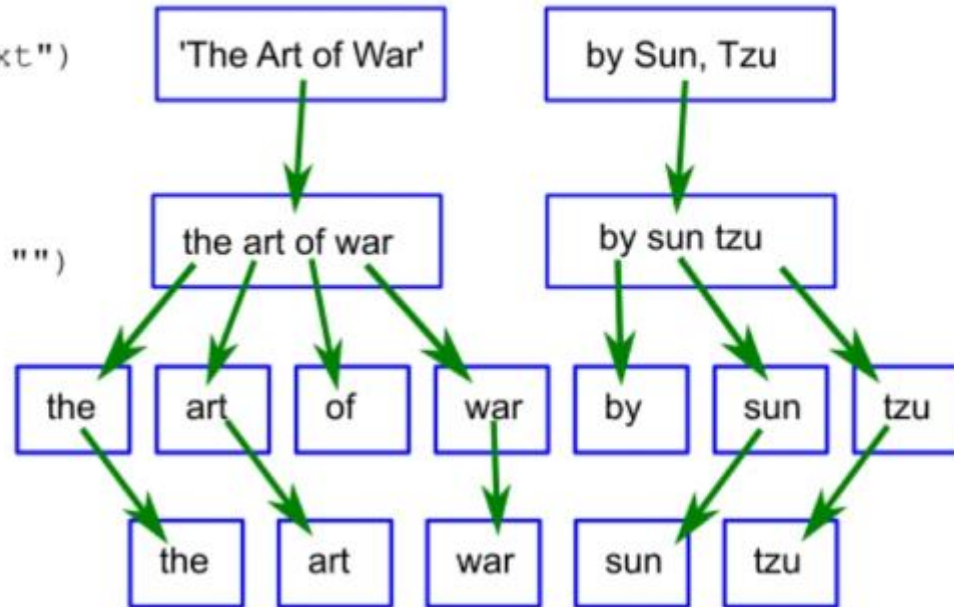
<http://sciabarra.com/blog/spark-visualizing-rdd>

```
sc.textFile("artofwar.txt")
```

```
.map(  
  _.toLowerCase  
  .replaceAll("[^\\w ]", "")  
)
```

```
.flatMap(_.split(" "))
```

```
.filter(_.size > 2)
```



I. Create a RDD – sc.parallelize

- See the RDD_example.ipynb

Cmd 2

```
1 myData={ (1,2), (3,4), (5,6), (7,8), (9,10) }  
2 print(myData)
```

```
{(1, 2), (7, 8), (5, 6), (9, 10), (3, 4)}
```

Command took 0.04 seconds -- by changs@ksu.edu at 6/30/2020, 2:23:59 PM on IMSE 785 (clone)

Cmd 3

```
1 #I. using parallize()  
2 #myData can be obtained through a database  
3 #rdd1 is a rdd by distributing the myData into worker nodes  
4 rdd1=sc.parallelize(myData)  
5 rdd1.collect()
```

► (1) Spark Jobs

```
Out[3]: [(1, 2), (7, 8), (5, 6), (9, 10), (3, 4)]
```

Command took 0.18 seconds -- by changs@ksu.edu at 6/30/2020, 2:25:32 PM on IMSE 785 (clone)

RDD Converted into a Dataframe: toDF

Cmd 4

```
1 # rdd can be converted into a dataframe
2 df1=rdd1.toDF(['col1', 'col2'])
3 df1.show()
```

► (5) Spark Jobs

►  df1: pyspark.sql.dataframe.DataFrame = [col1: long, col2: long]

```
+-----+-----+
|col1|col2|
+-----+-----+
|  1 |  2 |
|  7 |  8 |
|  5 |  6 |
|  9 | 10 |
|  3 |  4 |
+-----+-----+
```


II. Create a RDD – createDataFrame

- See the RDD example.ipynb

Cmd 6

```
1 #II. using createDataFrame() Note that this rdd is dataframe.
2 rdd2=spark.createDataFrame([ ('1', 'john', '1991'), ('2', 'mary', '2001') ],['ID','Name','Year'])
```

▶  rdd2: pyspark.sql.dataframe.DataFrame = [ID: string, Name: string ... 1 more fields]

Command took 0.11 seconds -- by changs@ksu.edu at 6/30/2020, 2:39:19 PM on IMSE 785 (clone)

Cmd 7

```
1 rdd2.take(2)
2 print(rdd2)
```

▶ (3) Spark Jobs

DataFrame[ID: string, Name: string, Year: string]

Command took 0.46 seconds -- by changs@ksu.edu at 6/30/2020, 2:42:53 PM on IMSE 785 (clone)

Cmd 8

```
1 #
2 rdd2.show()
```

▶ (3) Spark Jobs

```
+---+---+---+
| ID|Name|Year|
+---+---+---+
1  1|john|1991|
```

III. Spark Read and Load

Cmd 9



```
1 #III. spark read and load (this rdd is a dataframe)
2
3 file_location = "/FileStore/tables/flights_2015_summary-dbc99.csv"
4 file_type = "csv"
5 infer_schema = "true"
6 first_row_is_header = "true"
7 delimiter = ","
8
9 # The applied options are for CSV files. For other file types, these will be ignored.
10 rdd3 = spark.read.format(file_type) \
11     .option("inferSchema", infer_schema) \
12     .option("header", first_row_is_header) \
13     .option("sep", delimiter) \
14     .load(file_location)
15
16 display(rdd3)
17 # rdd3.show()
```

► (3) Spark Jobs

► rdd3: pyspark.sql.dataframe.DataFrame = [DEST_COUNTRY_NAME: string, ORIGIN_COUNTRY_NAME: string ...

IV. Use `sc.textFile` to read from HDFS

Cmd 10

```
1 #IV. use sc.textFile to read from hdfs
2 file_location = "/FileStore/tables/flights_2015_summary-dbc99.csv"
3 rdd4=sc.textFile(file_location)
4 #note that rdd4.show() won't work since it is not a dataframe
```

⊕ **AttributeError:** 'RDD' object has no attribute 'show'

Command took 0.11 seconds -- by changs@ksu.edu at 6/30/2020, 2:55:09 PM on IMSE 785 (clone)

Cmd 11

```
1 print(rdd4)
```

```
/FileStore/tables/flights_2015_summary-dbc99.csv MapPartitionsRDD[153] at textFile
a:0
```

Transformations and Actions

TRANSFORMATIONS

map

flatMap

filter

distinct

reduceByKey

mapPartitions

ACTIONS

collect

collectAsMap

reduce

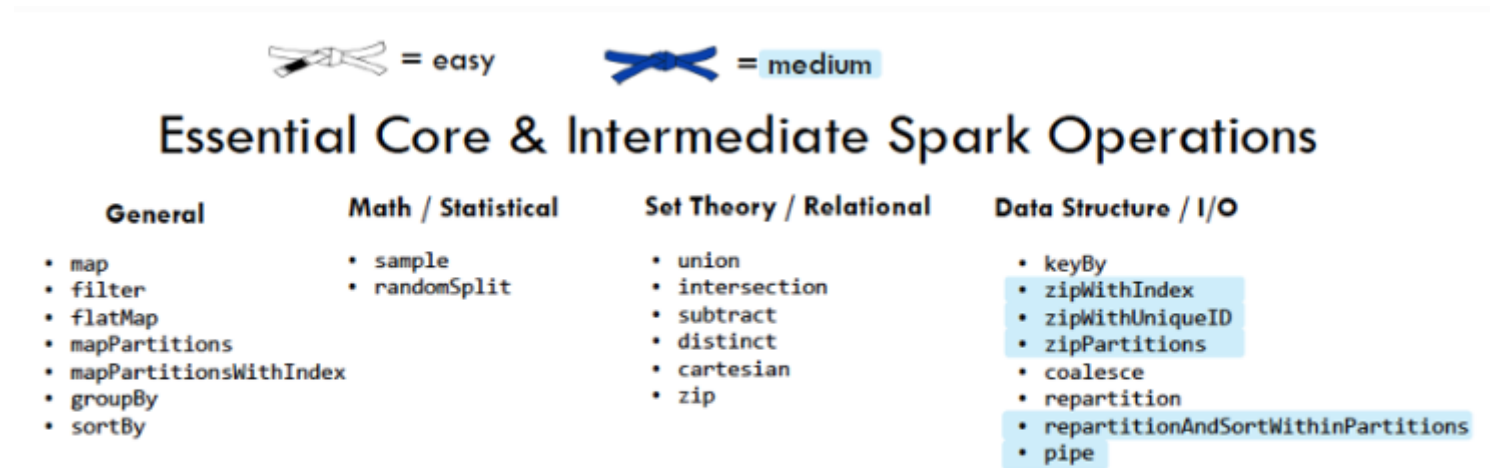
countByKey

take

countByValue

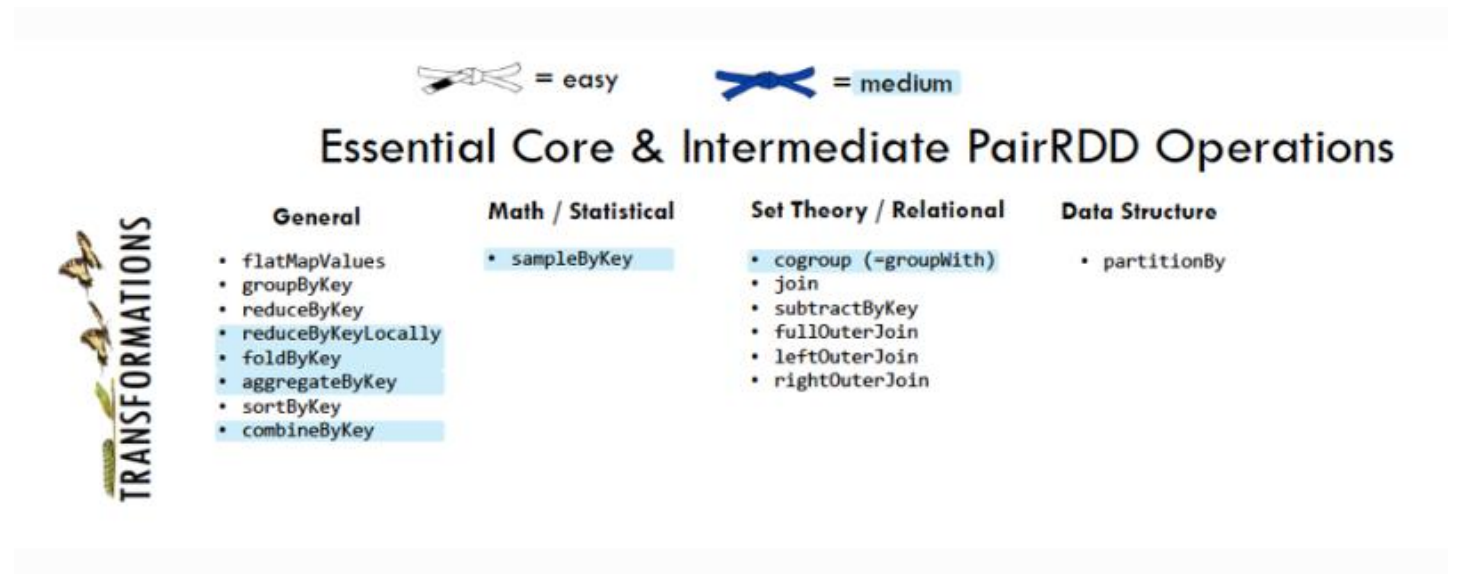
Transformations (1)

- Transformation
construct a new
RDD from a previous
one
- Examples:
 - map
 - flatmap
 - groupBy
 - distinct
 - filter
 - coalesce



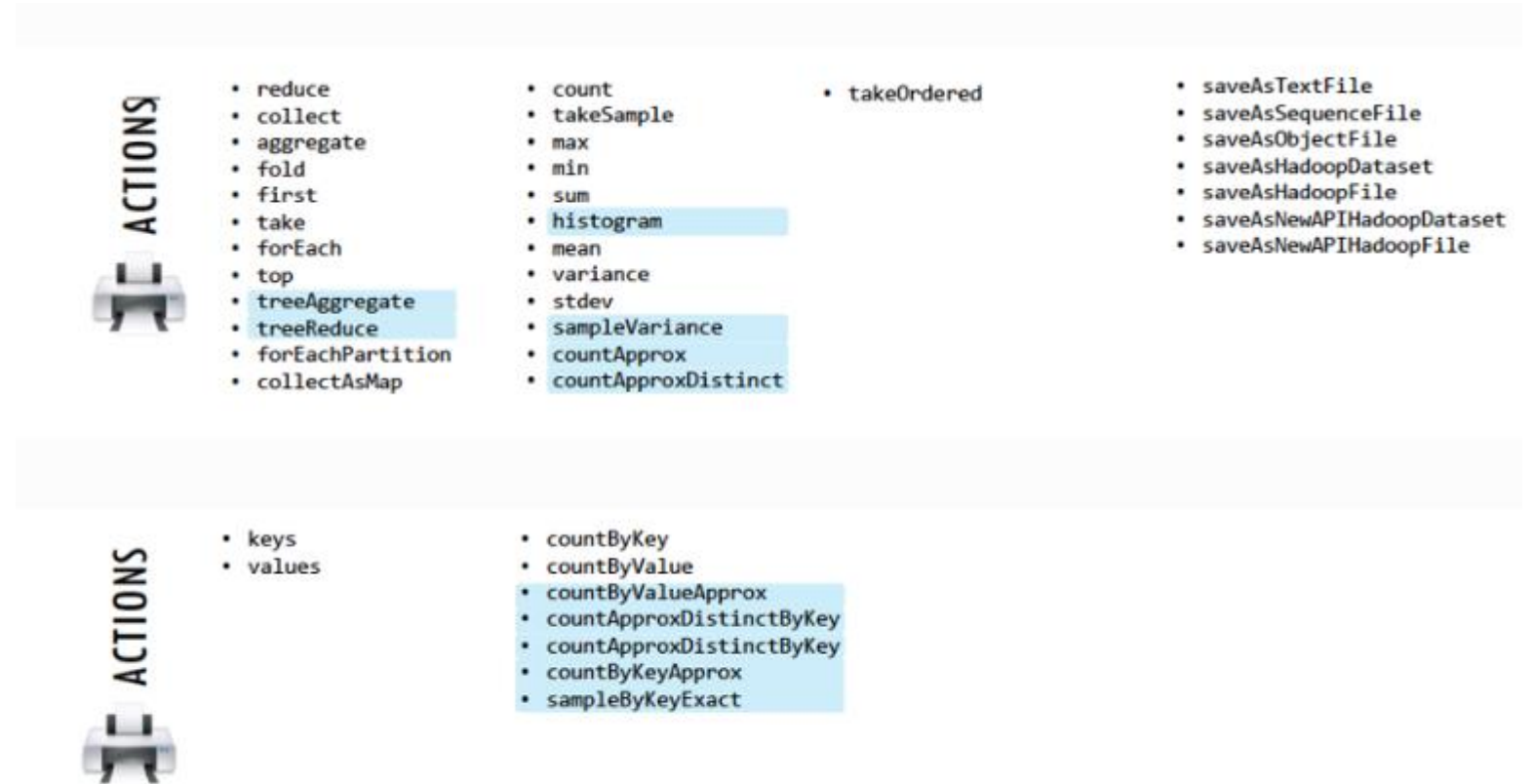
Transformations (2)

- Transformation construct a new RDD from a previous one
- Examples:
 - reduceByKey
 - groupByKey
 - sortByKey
 - flatMapValues
 - sampleByKey



Actions

- Actions compute a result based on an RDD
- Examples:
 - reduce
 - collect
 - take
 - top
 - count
 - max



Commonly used actions

- take()
- first()
- count() + countApprox()
- max() and min()
- saveAsTextFile()
- cache() data in memory +
getStorageLevel() in mem or disc
- reduce()

Your Turn

- Watch [PySpark RDD Tutorial](#) (about 32 minutes) but use your own datasets to experiment with various RDD transformations and actions
- Practice various commands in your own Databricks Community Edition Notebook (some codes may come in handy for creating your own codes)
- At about 13:47 mark, you will see a Wordcount code. Could you modify this code to get ride of the stopwords and punctuations using the Shakespeare txt?

Ref

- <https://github.com/bbengfort/hadoop-fundamentals/tree/master/spark>
- Edureka PySpark RDD tutorial: <https://youtu.be/e5ol7oyKV0A>
- Chambers and Zaharia, chapters 12 and 13